5.1 INTRODUCTION

The living organism performs so many vital processes. One of the processes where living organisms produce offspring of their own kind is called reproduction. Reproduction is the most fundamental function of living things. It is essential for continuing and survival of the species. An individual will not die if it does not reproduce but there is a danger of extinction, if does not reproduce regularly. First the species becomes endanger due to low rate of reproduction and finally vanish if not reproduce. Many species of plants and animals are in danger of vanishing because man use them or kill them in such large number which cross the limit of its rate of reproduction. Sometimes changes in the habitat of an organism due to the different activities, organism do not feel comfortable to reproduce. It means the reproduction also depends on the favorable environmental conditions also.

5.2 TYPES OF REPRODUCTION

Living organism can produce by two ways

(i) Asexually

(ii) Sexually

Asexual Reproduction	Sexual Reproduction
• Type of reproduction which takes place without fusion of male and female gametes(sex Cells).	• Types of reproduction which takes place by the fusion of male and female gamete (sex cells).
• In this type of reproduction only one parent is involved.	• In this type of reproduction usually two parents of opposite sexes are involved.
• The off springs are exactly similar to their parent.	• The off springs are not exactly similar to any one of the parent.
• Organisms are genetically similar to each other as well as to their parent.	• Organisms are genetically dissimilar to each other as well as to their parent.
• No new combination of genes (genetic recombination) occurs.	• Genetic recombination occurs which causes variation and leads to evolution.

5.2.1 Asexual Reproduction in Protists, Bacteria and Plants

The bacteria, protists and plants reproduce asexually by a number of methods, whereas Bacteria only reproduce by asexual reproduction.

Some of the asexual methods of reproduction in plants are described below.

(i) By Fission (Splitting):

The splitting of cell into two or more cells is called Fission. It is the simplest and the fastest mode of asexual reproduction during which there is replication of genetic material (Prokaryotes) or division of the nucleus (Eukaryotes) followed by division of cell (parent body) into independent daughter cells. Each daughter cell receives equal amount of nucleic or genetic material. Fission is of two types. i.e. binary fission or multiple fission.

(a) Binary Fission:

Type of fission where a mother cell divides into two daughter cells. It occurs during favorable conditions. It takes place in bacteria under favorable conditions of temperature, nutrition and moisture, single bacterium divides into two bacteria within 20 minutes and numerous bacteria are produced within very short interval of time.

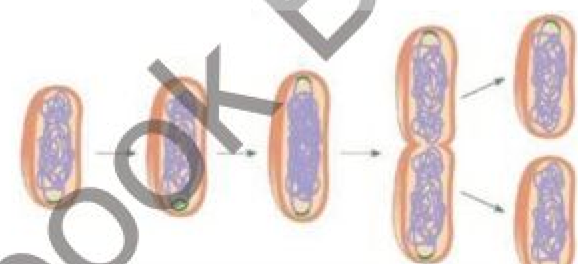


Fig. 5.1 Binary Fission

a) Multiple Fission:

Type of fission where a mother cell divide into more than two daughter cells

(ii) Budding:

In this type of asexual reproduction the parent cell forms a small out growth which is called bud. This bud detach from parent cell or body and grows into new organism. It takes place in yeast and plants.

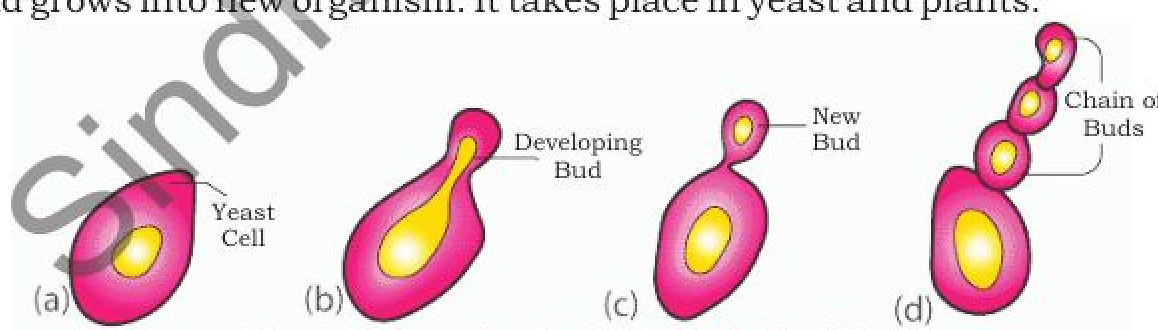


Fig. 5.2 Reproduction in yeast by budding

(iii) By Spores

In fungi, algae and plants asexual reproductive structure sporangium is developed on their body. These sporangia produce numerous unicellular spores. Spores are very small and light usually they dispersed by wind. Spores have thick, resistant walls which enable them to survive in unfavorable conditions. When these spores drop on proper substratum, they develop into new organism in favorable conditions.

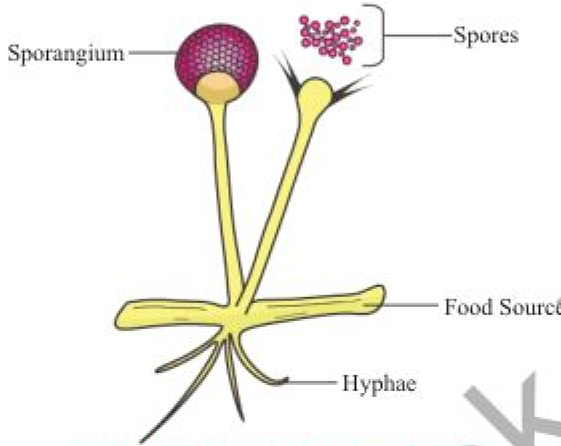


Fig. 5.3 Spore formation

(iv) Vegetative Propagation

Vegetative means non-reproductive part of plant like, thallus, root, stem and leaf. If any of this part develop into new plant called vegetative propagation. These parts sooner or later separated from parent plant. These specialized vegetative organs are morphologically different in different cases but usually in all cases they bear one or more bud which develops into new plants.

Runner (Grass and strawberry) stolen (mint), rhizomes (ginger)
Bulb (onion and garlic), stem tuber (potato), Bulbil (Bryophyllus)
Root tubers (sweet potato), Phylloclades (Opuntia)

The vegetative reproduction in most cases is due to accumulation of large amount of reserve food and consequent growth of the vegetative parts.

Fig. 5.4 Vegetative propagation in plants

Vegetative Propagation	Artificial Propagation
• It is natural development of new plant without human efforts. • Natural vegetative propagation usually occurs by root, stem, or leaves. • Stem; Runners grow horizontally above the ground. • Roots; new plant emerges out of swollen modified root known as tuber. • Leaves; of a few plants detached from parent plant and develop into new plant e.g. Bryophyllum.	• It is the method of development of new plant with the help of human efforts. • Artificial propagation can occur from cells, tissues, cutting of stem etc. • The method are Tissue culture, Cutting, Grafting, Layering, Budding • Root may be used for artificial propagation • Any leave tissue may also use for artificial propagation.

5.2.2 Vegetative Propagation in plants (through stem, suckers and leaves)

New plants can be produced from vegetative structures such as the roots, stems, suckers and leaves. The process can be natural or artificial.

By Stem:

In many plants, the stem has buds as in onions, daffodils and strawberries etc. have stems that can start new offspring. These type of stem that can reproduce recognized as runners, bulbs, rhizome, tubers, and suckers.

Fig. 5.5 Vegetative propagation by stem

Leaves:

Fig. 5.6 Bryophyllum

Some leaves have bud on their margin e.g. Bryophyllum. These buds give rise adventitious root when fall on ground or come in contact with soil. After some time these parts of leaves develop into an independent plant.

Suckers:

Fig. 5.7 Sucker

Suckers are known as root sprouts, basically these are plants stem that arise from buds on the base of stem or root of parent plant that use suckers, are apple, elm and banana tree. Suckers grow and form a dense compost mats that is attached to parent plant. Too many suckers can lead to a small crop.

5.5.3 Methods of Artificial Vegetative Propagation:

Plants have unique feature that they possess growing or embryonic centers in the form of bud. Due to these embryonic centers, plants are capable of farming new plants. The man has exploited it to reproduce desired type of plants by artificial propagation. It takes place by cutting, grafting and cloning.

Cutting:

Fig. 5.8 Cutting of Sugarcane

Cuttings are the short pieces of stems that have 2 to 3 nodes and buds. It cuts obliquely below a node from a plant. The cutting embedded in soil with at least one node above the soil. The adventitious roots and shoots grow from buds of the portion below the soil and above the soil respectively forming a new plant e.g. sugar cane, sweet potato and rose.

Grafting:

This is a technique where a branch of desired variety of plant is joined to another plant with well-established root system. The plant from which the branch is taken is called **scion** and the plant to which it is joined is called **stock**. The two plants involved are normally belongs to different varieties of same species e.g. oranges, lime and mango.

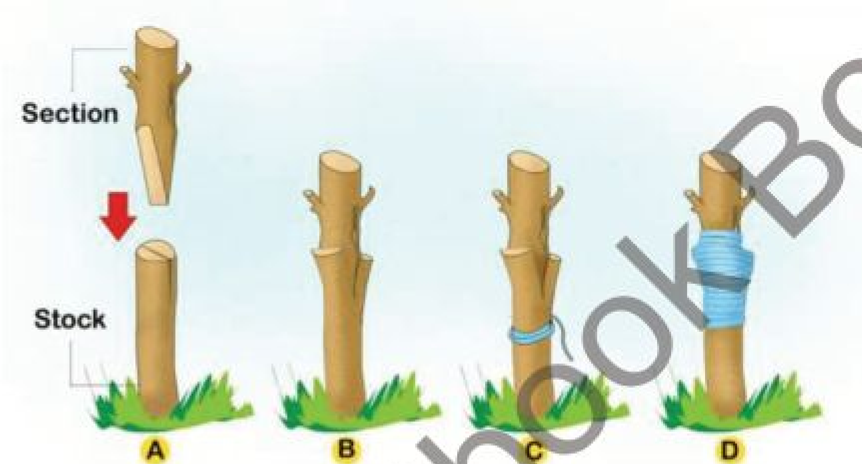


Fig. 5.9 Grafting of plant

Cloning (Tissue Culture Technology)

It is one of the recent techniques. In this method tissues of selected plants are cultured using their ability of asexual reproduction in test tube or dishes. To speed up the growth, hormones are added in the growth medium. After some time the baby plant is transferred to field for large commercial scale production of desired yield.

5.2.4 Apomixes (Parthenogenesis) is the type of asexual reproduction:

Apomixes is the type of seed production without fusion of male and female gametes. It is the type of Parthenogenesis which is the form of asexual reproduction. Where an egg develops into a complete individual without being fertilized. The resulting offspring can be either haploid or diploid depending on the process and the species. It is a natural form of asexual reproduction in which embryo develops in the absence of fertilization. Most commonly found in plants and invertebrates. The question arises here when gamete is involved so why it is considered as asexual reproduction? Because this process does not require fusion of male and female gametes to produce offspring and new genetic combination does not occur.

5.2.5 Sexual Reproduction in flowering Plants

The angiosperm is the group of plants which gives rise traditional flower therefore these plants are called flowering plant. In these plants sexual reproduction takes place through flower. Flower is highly modified shoot which is responsible for the reproduction by producing seeds within fruits.

Angiospermic flower has two external wheels of leave called **calyx** and **corolla** which consist of **sepals** and **petals,** respectively. The **androecium** and **Gynoecium** are two inner wheels of leaves i.e. **stamen** and **carpels** respectively; which are directly responsible for sexual reproduction of plants. Stamen produce pollen **grains** and carpel produce **ovule** in ovary.

Fig. 5.10 Parts of a flower

Structure of Ovule:

Each ovule has main cellular body called **nucellus**. It is surrounded by two coats, outer and inner integuments. A small opening present at the apex of the integument is called **micropyle**. The ovule has stalk, the **funicle** with which it is attached to ovary wall. **Chalaza** are tissues between nucellus and funicle.

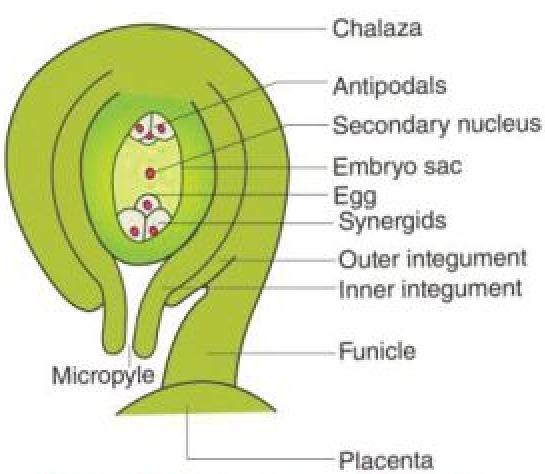


Fig. 5.11 Structure of ovule

There is a large oval cell embedded in the nucellus which form embryo sac (female gametophyte). The mature embryo sac consists of 7 cells i.e. one Ovum, two synergids, three antipodals cells and one secondary nucleus in numbers while a diploid cell is a fusion nucleus in the center.

Structure of pollen grain:

Pollen grain develops in pollen sac of anther form microspore in the form of loose, dusty powder. Each pollen grain is 4 celled structure, bounded by wall which consists of 2 layers, outer, the **exine** and inner, the **intine.**

In angiospermic plants the main plant is sporophyte which consists of **vegetative** and **floral** parts. The vegetative parts are root, stem and leaves while flower, fruit, seed are floral parts which develop from flower. The floral part is reproductive part. After two processes i.e. pollination and fertilization, it produces seed within fruit. The seeds when disperse germinate into baby plant called **seedling**, when matures, it become a new plant like its parents.

In flower androecium (male part) is consist of stamen (microsporophyll) has 2 to 4 pollen sac (microsporangia) in its anther. These pollen sac are filled with microspore mother cell which produce microspore by meiotic cell division. Each unicellular microspore divide its cells by mitosis and produce 2 to 4 cells, in this way unicellular microspore become pollen grain (multicellular) but each of its cell is haploid. When anther burst these pollen grains disperse in nature.

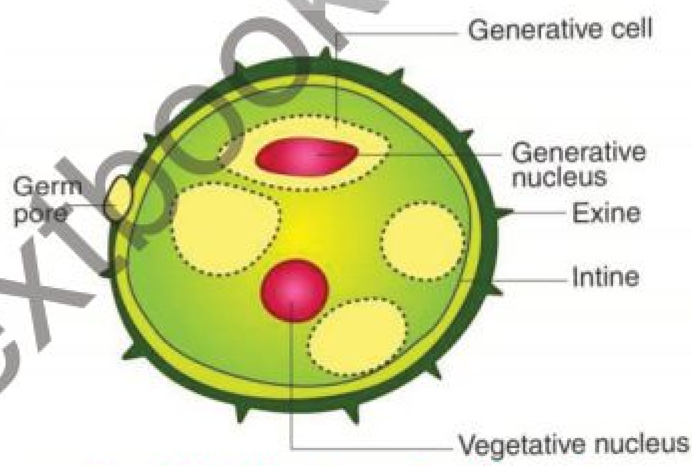


Fig. 5.12 Structure of pollen grain

On the other hand each carpel (megasporophyll) has one or more ovules (megasporangium) in its ovary. Each ovule has single megaspore mother cell. This megaspore mother cell divided by meiosis to produce 4 haploid megaspores. Only one will survive and develops into embryo sac (female gametophyte) inside ovule, which consists of 7 cells as discussed earlier. The pollen grain disperse if dropped at the stigma of carpel the life cycle remain continues. Now the question arise how pollen grain transferred to stigma and what is it called.

5.3 POLLINATION

Pollination is the process in which pollen grains are transferred from anther to stigma of carpel.

5.3.1 Types of Pollination

(i) Self pollination
(ii) Cross pollination

(i) Self Pollination

It is the transfer of pollen grains from anther of stamen to the stigma of same flower or flowers on same plant.

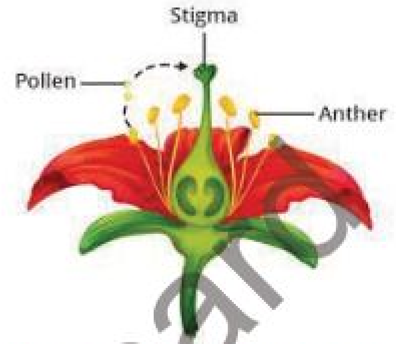


Fig. 5.13 Self pollination

(ii) Cross Pollination

It is the transfer of pollen grains from anther of one flower to stigma of other flower belongs to another plant of same species.

Cross pollination is more common than self-pollination, the pollen grains are carried from one flower to another flower through following agents.

(i) Wind
(ii) Water
(iii) Insects
(iv) Animals

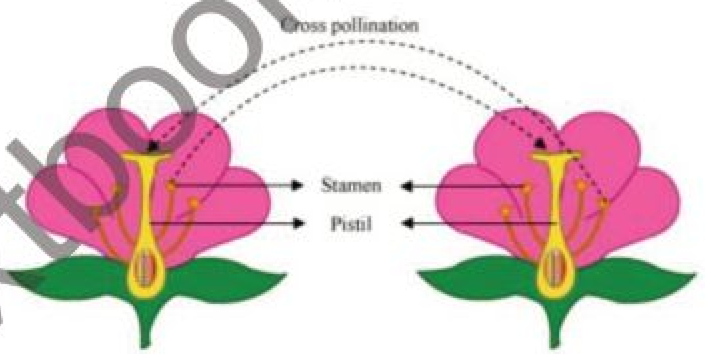


Fig. 5.14 Cross pollination

When pollen grain drops at stigma, it starts its development into pollen tube (male gametophyte), which consists of 6 haploid cells, among them two are prothallial cells, two are male gametes, one stalk nucleus and one tube nucleus. The pollen tube grows from stigma to ovule through style and transfers two male gametes in ovule through micropyle which ultimately reach to embryo sac. One sperm nucleus fuses with ovum to produce diploid (2N) zygote while other gamete fuses with secondary nucleus to form 3N (Triploid) cell which later develop into endosperm of seed. This type of fertilization is called **double fertilization** which is the characteristic feature of angiospermic plant.

The 2N zygote after successive mitotic divisions develops into an **embryo** within the embryo sac, whereas triploid (3N) secondary nucleus develops into **endosperm**. The endosperm provides nourishment to the developing embryo.

During this development the ovule develops into **seed** the integument develop into **seed coat** whereas zygote form small embryo and cotyledon during this the ovary outside ovule become swollen due to mitotic cell division and become **fruit**. The fruit is eaten by animal or decay, the seeds come out, disperse or dropped in soil. In favorable conditions, it germinates and grows into new baby plant

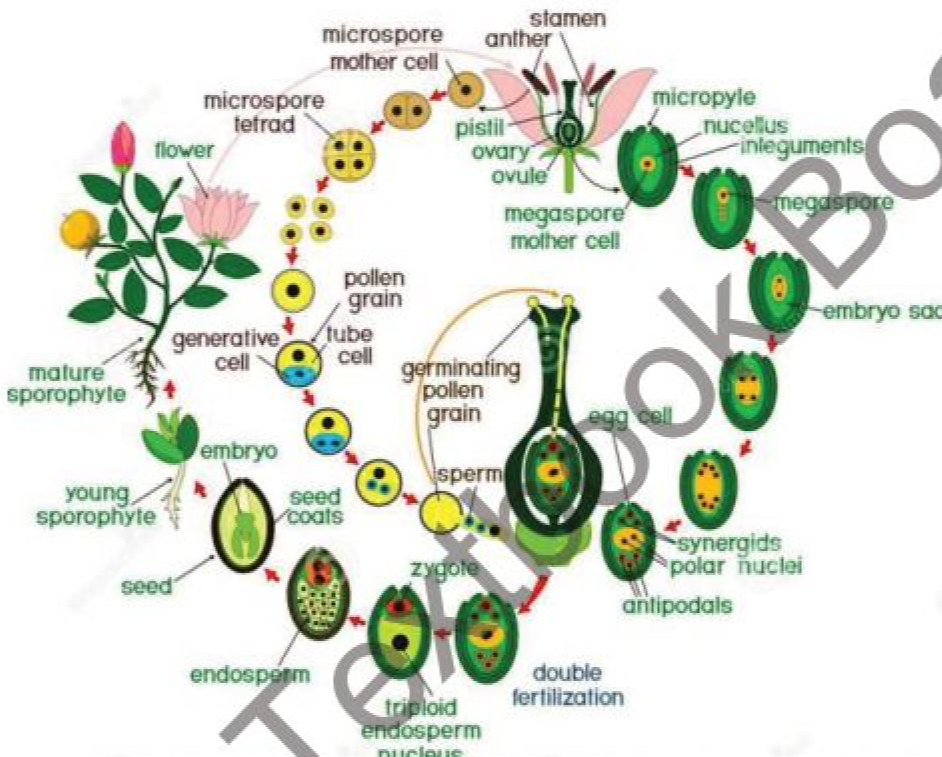


Fig. 5.16 Life cycle of flower

Fruit Formation

The ovary enlarges to form a fruit, containing seed or seeds. The stigma and style disappear. The stamens and petals are usually lost after pollination. In some cases, sepals remain attached with fruit e.g. brinjal. The endosperm provides nourishment to develop embryo. It also stores energy in grains like wheat, rice, gram etc. which can be utilized by us or other animals.

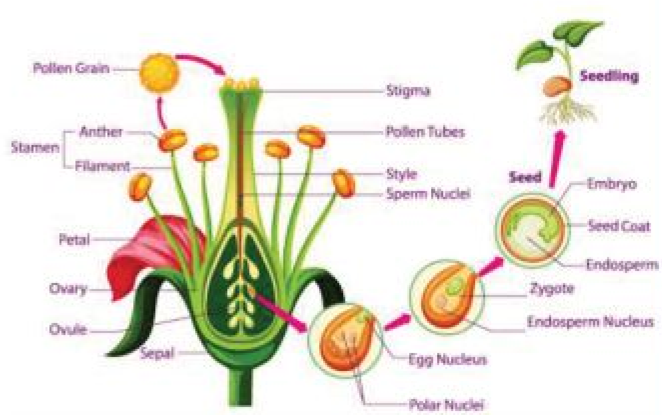


Fig. 5.16 Fruit formation

Formation of fruit with out fertilization

Fruit and seed formation usually occur after fertilization but some fruit may formed without fertilization. This mechanism is called **parthenocarpy** i.e formation of seedless fruit, like banana.

5.3.2 Adaptation in the structure of wind pollinated and insect pollinated flowers.

Some plants spread their pollen grains by wind and H_2O while the other spread by insects and animals. The plants which spread their pollen grain by wind and H_2O have some adaptive characters.

Adaptive characters of wind and water pollinated plants.

1. The flowers of these plants are non- attractive, small in size and do not bear any odour.
2. They produce pollen grains in high quantity.
3. The pollen grains are very light in weight, some of them bear wings and some have parachute like structure.
4. They do not produce high quantity of nectar.

Adaptive character in insect pollinated plants.

1. The flowers are large in size.
2. They have bright coloured petals or sepals or bracts.
3. The pollen grains have sticky substance or hooks.
4. They produce special odour
5. They produce high quantity of nectar.

5.3.3 Seed and its structure

Seed may be defined as ripened ovule or it is fertilized, developed ovule which contains dormant embryo.

The seed consists of following parts:

1. Seed Coat
2. Embryo
3. Cotyledon
4. Sometime endosperm

- The outer wall of seed which develops from integument of ovule called seed coat. The seed coat consists of an outer thick layer called **testa** and inner thin wall called **tegmen**. The embryo develops from diploid zygote. It is a small axis, lying between two cotyledons, the upper end called **plumule** and the other lower end

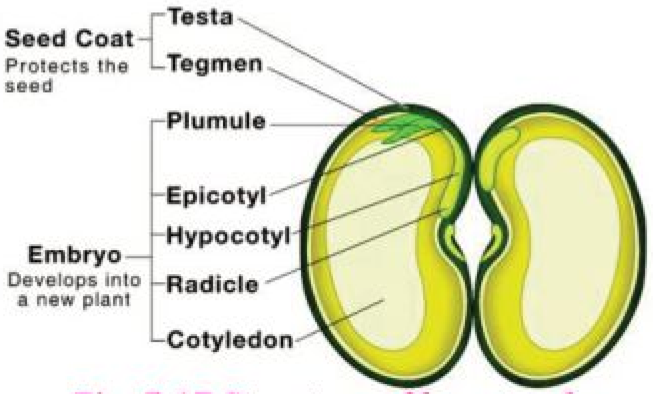


Fig. 5.17 Structure of bean seed

is called **radicle**, during germination plumule develops into shoot while radicale develops into root. The seed also contains leaf-like structure called **cotyledon**. These are either one or two on the basis of these numbers the seeds are classified into **monocot** or **dicot** seed, respectively. In endospermic seed these cotyledons are thin and paper like. In many of the seeds endosperm is not present than food is stored in cotyledons therefore they become swollen and thick e.g pea. The hilum is a scare, present at seed coat. The water enters into the seed through a very small hole in the seed coat this pore is called **micropyle**.

In some monocot seeds, ripened ovary walls called **pericarp** get fused permanently with seed coat as found in maize grains. Internally maize grain is divided into two unequal parts by a thin layer of cells called **epithelium**. The larger portion is t
he **endosperm** and the smaller is **e**
mbryo. In the embryonic part, a s
hield shaped cotyledon is present c
alled **scutellum**. Moreover the p
lmule and radicle are enclosed in p
rotective sheath called **coleoptile** a
nd **coleorhiza,** respectively.

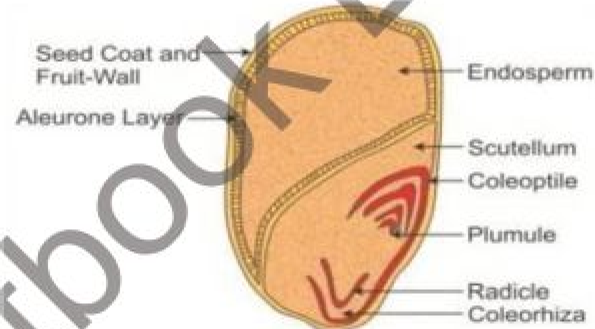


Fig. 5.18 Structure of maize grain

5.3.4 Germination of seed

Breaking of seed dormancy is called seed germination. As a result of germination seed develops into seedling.

Conditions necessary for seed germination

Only living seeds can germinate, require optimum condition of moisture, oxygen and temperature.

(i) Role of water (Moisture)

Water is essential for life because metabolic activities depend on water. Seed coat become soft by water. Cotyledons and endosperm absorb water by imbibition become swollen and exert pressure on seed coat to break. So the embryo comes out to grow, enzymes become activated by water and solid reserve food change into solution.

(ii) Role of oxygen:

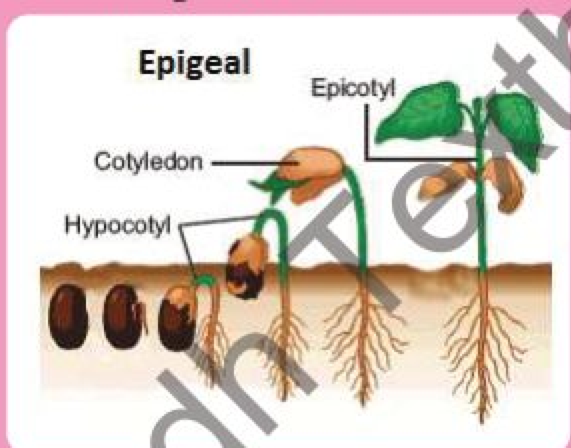
The metabolic activities require energy. Energy is produced during respiration which requires oxygen.

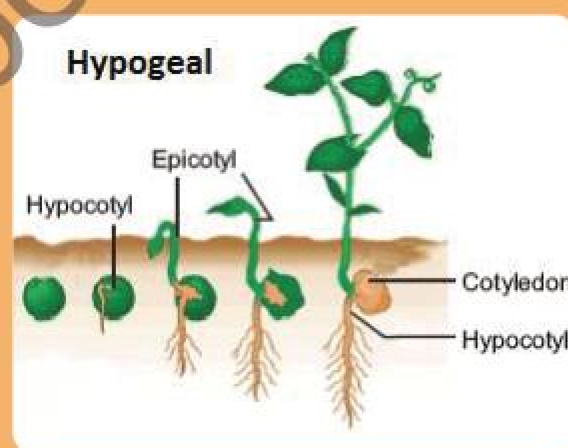
(iii) Temperature:

Enzymes activity require certain range of temperature. Most of the seeds require the temperature range between 25 to 37°C. Seeds do not germinate at temperature below 0°C or above 45°C.

Types of germination

Epigeal Germination	Hypogeal Germination
Epi = above, geo = earth The type of germination where seeds come above the soil during germination.	Hypo = below, geo = earth The type of germination where seeds remain in the soil during germination
The growth rate of hypocotyl is higher than epicotyl.	The growth rate of epicotyl is higher than hypocotyl.
The hypocotyl grow in the form of arch.	The hypocotyl does not grow in the form of arch.
The cotyledons become green when come above the soil and work as 1st foliage leaves.	The cotyledons do not turn green.





5.4 REPRODUTION IN ANIMALS

There are also two types of reproduction in animal. i.e. Asexual and sexual reproduction.

5.4.1 Asexual reproduction

Animals reproduce asexually by different methods, some of them are as follows.

(i) Fission (Splitting)

Splitting of cell into two or many cells or organisms called fission.

(a) Binary Fission:

The type of fission where an organism divides into two organisms is called binary fission. It is commonly observed in unicellular organisms like protozoa. During this process the nucleus of the parent organism divides into two nuclei, both of them move in opposite directions in the cytoplasm. Meanwhile, a constriction appears in cytoplasm which deepens from outside to inside finally organism divides into two organisms.

(b) Multiple Fission

Multiple fission involves the division of an organism into many small sized daughter organisms as found in *Plasmodium*.

(ii) Budding

In this method one or more out growth develop on the body surface of organism which are called buds. When buds separates from the parent body starts living independently and develop into new organism e.g. *Hydra*.

(iii) Fragmentation

It is found in lower, multicellular animals like liver fluke and nematodes. When a living organism divides into fragments, each fragment recovers its lost part by regeneration and develops into new organism.

5.5 SEXUAL REPRODUCTION

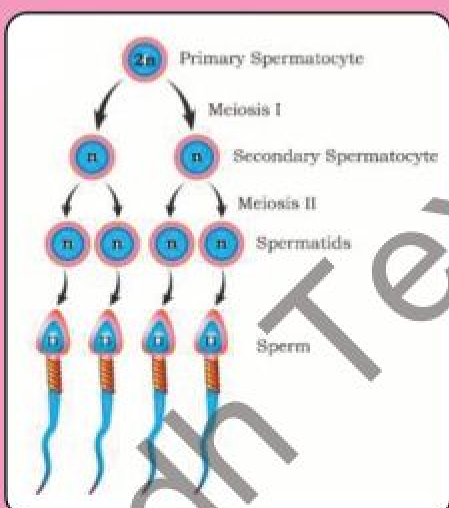
The process of sexual reproduction involves fusion of specialized haploid sex-cells or gametes to form a single diploid cell, **zygote**. The fusion of these gametes is called **fertilization**. Sexual reproduction involves:

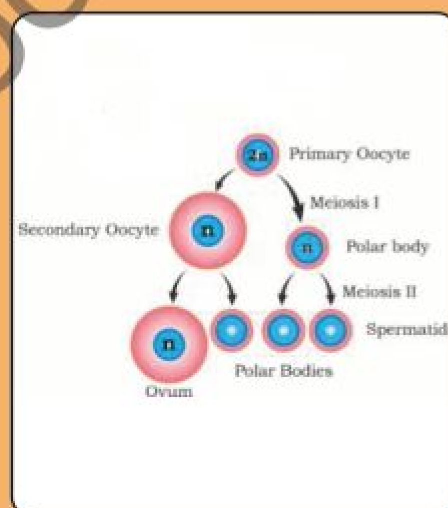
(i) Gametogenesis - Formation of gametes
(ii) Mating - Union of male and female organisms to collect their gametes at same place.
(iii) Fertilization - Fusion of male and female gametes to form zygote.

5.5.1 Gametogenesis in Rabbit:

Gametogenesis is the process of gametes formation by meiosis in gonads.
These are two types.

Spermatogenesis	Oogenesis
Formation of sperm in male gonads (Testis) from germ cells.	Formation of ovum in female gonads (ovary). The process of gametes formation or gametogenesis involves meiosis, which reduces number of chromosomes to half in gametes. It also results variation in genes by process called **crossing over** .These gametes mostly do not have identical genetic make up.





5.5.2 Male and Female reproductive Organs in Rabbit

	Male	Female
Gonads (Gametes producing organs)	• Testes two in numbers • Hanging outside, in a sac called **scrotal sac**. • Contain seminiferous tubules • Epididymis to collect sperm	• Ovaries two in numbers • Located in abdominal cavity • Produce ovum
Duct (Gametes collecting tubes)	• Vas deferens, two in number.	• Oviduct or fallopian tube collect ovum from ovary. • Two in number
Genitals (Gametes depositing or receiving organs)	• Penis, a muscular organ to transfer semen into female genital	• Vagina, a tube which receive semen containing many sperms
Glands	• Prostate gland • Cowper's gland • Seminal vesicle	• Ovary works as gland as well

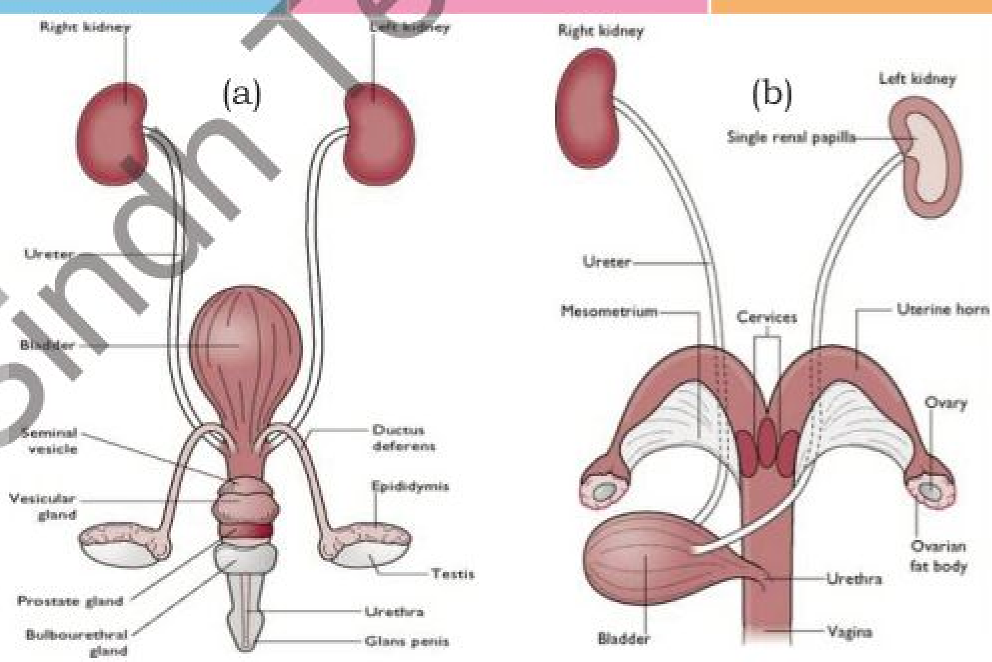


Fig. 5.19 Reproductive organs of rabbit (a) Male (b) Female

5.5.3 Fertilization

The process of fusion of male and female gametes from diploid zygote is called **fertilization**. On the basis of location of fertilization there are two types of fertilization, **external** and **internal** fertilization.

External Fertilization	Internal Fertilization
• It takes place outside the body.	• It takes place inside the body of female.
• It takes place in water.	• It takes place inside female body.
• Both gametes mature at same time.	• Gametes may mature after each other.
• Gametes are produced in large numbers.	• Gametes are produced in limited numbers.
• It takes place in fishes and amphibians.	• It takes place in reptiles, aves and mammals.

5.6 NEED OF POPULATION PLANNING

Population planning is the policy to limit the growth in number of population especially in those countries which has limited resources or in densely populated part of earth.

Population means the total number of beings living in a particular area. The human population helps in estimating the birth and death rates, the number of resources that will be required i.e. food, houses, health, electricity, transport, drinking water, garbage disposal etc.

In the modern world people wants to live quality life, better health facilities and jobs. The average is higher than before which is leading to an increase in the population. An increase in population can cause strain on the resources due to high demands. This strain can lead to environmental disturbance.

To establish and maintain the quality of human life and environmental conditions the birth rate should be planned according to available resources of the area. It is necessary that the birth and death rates balances each other to maintain sustainable population growth. China and India are considered as overpopulated countries. It means that their population exceeds their caring capacity resulting in depletion of resources and threatening environment. United Nations and other world organisations formulated certain policies and strategies to keep a check on increasing population.

5.7 SEXUALLY TRANSMITTED DISEASES

The diseases or infections which are passed from one person to another person through genital organs and genital fluids during sexual contacts called **sexually transmitted diseases**. Sometimes they can spread through intimate physical contact as well like herpes, spread by skin to skin contact. Some of the sexually transmitted diseases are Gonorrhea, AIDS, Syphilis, Genital herpes etc.

Sexually transmitted infections have been known since ancient time, they remain world wide a major public health problem. AIDS is one of the serious threat came into light around 1980.

AIDS (Acquired Immuno Deficiency Syndrome) is caused by virus known as HIV (Human Immuno-deficiency Virus). The HIV is transmitted through contaminated surgical instruments, transfusion of infected blood, sexual contacts, placenta and mother's milk.

Role of national AIDS Control Program and different NGOs in educating about AIDs.

The first role was the study of AIDS patients, frequency in different social group of population. The next goal was the study of reasons of AIDS in different populations. Another aim was to educate people about prevention. The next role is the proper diagnosis of HIV in different private and government sector hospitals.

SUMMARY

- Reproduction is the vital process by which living organisms produce off springs of their own kind.
- Reproduction is of two types
  (a) Asexual (b) Sexual Reproduction
- Asexual reproduction takes place without fusion of gamets i-e no genetic recombination.
- Sexual Reproduction takes place as a result of male and female gametic fusion i-e Genetic recombination occur.
- Asexual reproduction in Protist, Bacteria and Plants takes place by fision, budding, spores, vegetative propagation.
- Vegetative natural propagation means reproduction takes place by any part of plant except flower i-e by roots, stem, leaves and sucker.
- Artificial vegetative propagation takes place by cutting, grafting, clonning and apomixes.
- Sexual reproduction requires flower where stamen produce pollon grain which develop into male gametophyte i-e pollen tube, where as carpel contain ovule in its ovary.
- Ovule has embryo sac (female gametophyte) which produce ovums.
- Pollination is the transfer of pollen grains from anther to stigma of carpel.
- There are two types of pollination i-e self and cross pollination.
- After pollination pollen grain develop pollen tube carry male gamets to ovule where ovum is present.
- One of the male gamet fuse with ovum to produce 2N Zygote and 2N Secondary nucleus fuse with another male gamet to produce 3N Zygote. i-e Double fertilization.
- The ovule after fertilization develop into seed where as ovary develop into fruit.
- Seed is fertilized develop ovule contain dormaint embryo.
- The breaking of seed domaracy is called germination.
- Seed germinate into two ways i-e Epigeal and Hypogeal germination.
- Asexual reproduction in animal takes place by fission, budding, fragmentation (Regeneration).
- Sexual reproduction has three steps
  i) Gametogenesis ii) Mating iii) Fertilization.

- Spermetogenesis is the formation of sperm, where dogenesis is the formation of ovum.
- Male and female reproductive organs are
  - i) Gonades to produce gamets
  - ii) Ducts; gamets collecting tubes
  - iii) Genitals; gamets donating or receiving organs.
- Glands of male are prostrats, cowper's and seminal vesicle.
- Glands of female are ovaries.
- Fertilization is the fusion of male and female gametes form diploid zygote (2N)
- There are two types of fertilization i-e external and internal fertilization.
- Population planning is a policy to limit the growth in number of population.
- Sexually transmitted disease are those which are passed from one person to another through genital organ and genital fluid.

EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The process which is essential for continuing and survival of species is

(a) Digestion (b) Respiration
(c)Reproduction (e)Excretion

ii) The type of reproduction which is necessary for evolution is

(a) Vegetative propagation (b) Fragmentation
(c) Sexual reproduction (d) clonning

iii) The unicellular structure, responsible for asexual reproduction without fusion is

(a) Pores (b) Spores
(c) Gametes (d) Pollen grains

iv) The example of stem which run horizontally on surface of soil to produce vegetatively

(a) Mint (b) Ginger
(c) Onion (d) Bryophyllum

v) Plant stem that arise from buds on the base of parent plants are

(a) Bulb (b) Rhizome
(c) Sucker (d) Runner

vi) The type of seed production without fusion of male and female gametes is
(a) Parthenocarpy (b) Apormixes
(c) Grafting (d) Scion

vii) The female gametophyte of anglosparmic plant is
(a) Embryo sec (b) Ovule
(c) Ovary (d) Carpel

viii) The 3N zygote is angiosperm develop into
(a) Seed coat (b) Cotyledon
(c) Embryo (d) Endosperm

ix) The male gonades in rabbit are
(a) Testis (b) Ovaries
(c) Scrotal sec (d) Vasdeferens

x) The female gametes are fertilized in the rear end of
(a) Oviduct (b) Follopin tube
(c) Ovaries (d) Both a and b

B. SHORT QUESTIONS:

i) Distinguish between asexual and sexual reproduction, Epigial and Hypogial Germination.
ii) Draw neat and labeled diagram of T.S of Angiospermic flower.
iii) How a new plant develops with an already growing plant?
iv) How leaves develop into new plants?
v) Draw neat and labelled diagram of Ovule.
vi) Draw neat and labelled diagram of male gametophyte of angiospermic plant?
vii) List out the male reproductive organs of rabbit with glands.
viii) What are STD's?
ix) Why population control is considered important for prosperous society?
x) Draw life cycle of Angiospermic plant.

C. EXTENSIVE RESPONSE QUESTIONS:

i) Describe natural asexual reproduction in plants.
ii) What is pollination? give adaptive characters of wind and insect pollinated flowers.
iii) Describe different types of asexual reproduction in animals.
iv) Describe the process of Spermetogenesis.
v) What is germination? Give the condition of germination also describe the methods of different germination.

Chapter 6

INHERITANCE

Major Concept

In this Unit you will learn:

- Introduction
- Chromosomes and Genes
- Law of Segregation
- Law of Independent Assortment
- Variation and Evolution